

DATA WAREHOUSING & DIMENSIONAL MODELING

Dimensional Modeling

- Data modeling for Data Warehouses
- Based on
 - Fact tables
 - Dimension tables

Fact Tables

- Represent a business process
- Contain the measurements or metrics or facts of business processes
 - "monthly sales number" in the Sales business process
 - most are additive (sales this month), some are semi-additive (balance as of), some are not additive (unit price)
- the level of detail is called the “grain” of the table
- contain foreign keys for the dimension tables

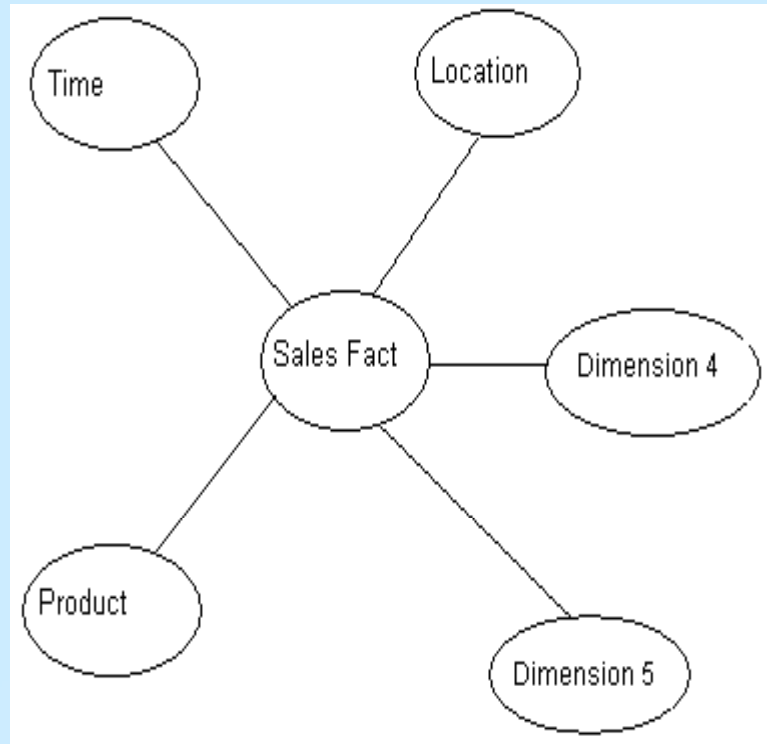
Dimension Tables

- Represent the who, what, where, when and how of a measurement/artifact
- Represent **real-world entities** not business processes
- Give the context of a measurement (subject)
- For example for the Sales fact table, the characteristics of the 'monthly sales number' measurement can be a Location (Where), Time (When), Product Sold (What).
- The **Dimension Attributes** are the various columns in a dimension table.

Dimension Tables

- In the Location dimension, the attributes can be Location Code, State, Country, Zip code.
- Generally the Dimension Attributes are used in report labels, and query constraints such as where Country='USA'. The dimension attributes also contain one or more hierarchical relationships.
- Before designing your data warehouse, you need to decide **what this data warehouse contains**. Say if you want to build a data warehouse containing monthly sales numbers across multiple store locations, across time and across products then your dimensions are: Location, Time and Product

Star Schemas



Advantages:

- easy to understand
- better performance
- extensible

Select the measurements

SELECT P.Name, SUM(F.Sales) . . .

JOIN the FACT table with Dimensions

**FROM Sales F, Time T, Product P,
Location L**

**WHERE F.TM_Dim_Id = T.Dim_Id
AND F.PR_Dim_Id = P.Dim_Id
AND F.LOC_Dim_Id = L.Dim_Id**

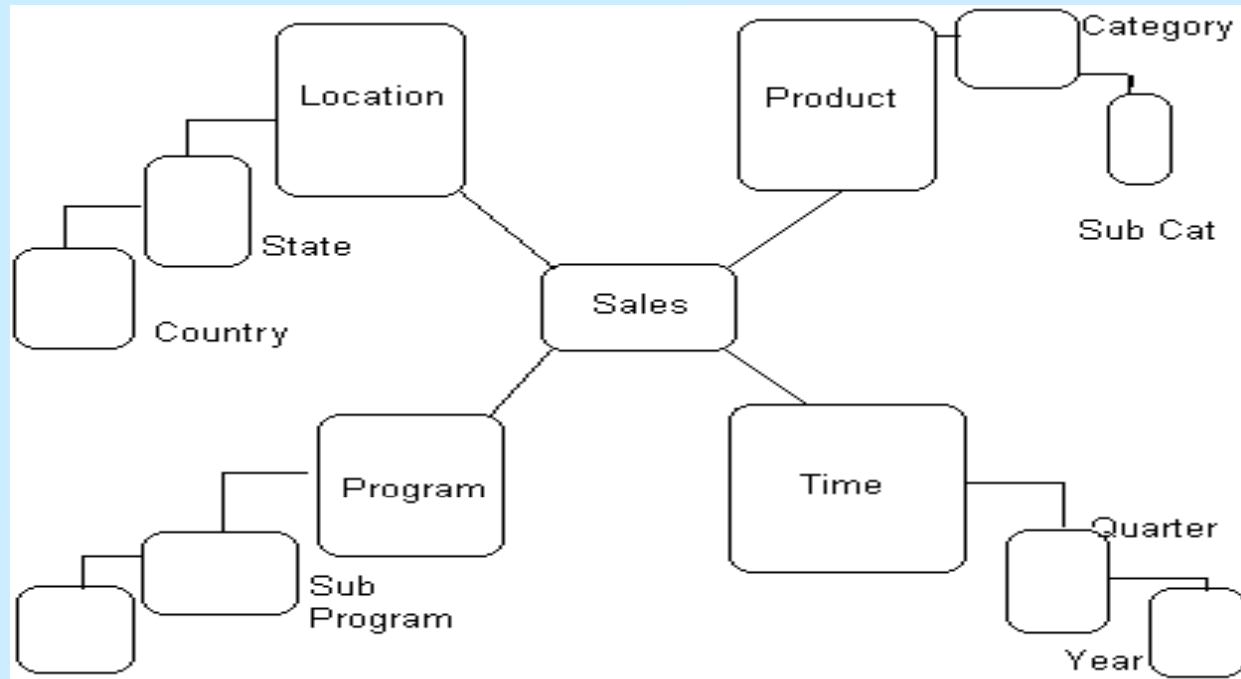
Constrain the Dimensions

**AND T.Month='Jan' AND T.Year='2003' AND
L.Country_Name='USA'**

'Group by' for the aggregation level

GROUP BY P.Category

Snow-flake Schemas



If we did not de-normalize
the dimensions

Rule of thumb:
don't use them

Four-Step Dimensional Design Process

- Objective: design of a dimensional database by considering four steps in a particular order:
 1. Select the business process to model
 2. Declare the grain of the business process
 3. Choose the dimensions that apply to each fact table row.
 4. Identify the numeric facts that will populate each fact table row

(Presentation based on Ralph Kimball, Margy Ross - The Data Warehouse Toolkit, Second Edition, Wiley & Sons, 2002, pp.29-65)

1. Select the business process

- A process is a natural business activity performed in the organization
- Typically is supported by a source data-collection system.
- Example business processes include:
 - raw materials purchasing,
 - orders,
 - shipments,
 - invoicing,
 - inventory,
 - general ledger

1. Select the business process – cont.

- It is not an organizational business department or function.
- For example, we build a single dimensional model to handle orders data rather than building separate models for the sales and marketing departments, which both want to access orders data.

1. Select the business process – cont.

- If we establish departmentally bound dimensional models, we'll inevitably duplicate data with different labels and terminology.
- Multiple data flows into separate dimensional models will make us vulnerable to data inconsistencies.
- The best way to ensure consistency is to **publish the data once**.
- A single publishing reduces the extract-transformation-load development effort

2. Declare the grain

- Declaring the grain means specifying *exactly what an individual fact table row represents*.
- The grain conveys the level of detail associated with the fact table measurements.
- It provides the answer to the question, “How do you describe a single row in the fact table?”

2. Declare the grain – cont.

Example grain declarations include:

- An individual line item on a customer's retail sales ticket as measured by a scanner device
- A line item on a bill received from a doctor
- An individual boarding pass to get on a flight
- A daily snapshot of the inventory levels for each product in a warehouse
- A monthly snapshot for each bank account

2. Declare the grain – cont.

- An inappropriate grain declaration will haunt a data warehouse implementation.
- Declaring the grain is a critical step that can't be taken lightly.
- If in steps 3 or 4 we see that the grain statement is wrong we must return to step 2, redeclare the grain correctly, and revisit steps 3 and 4 again.

3. Choose the dimensions

- If we are clear about the grain, then the dimensions typically can be identified quite easily.
- Represent all possible descriptions that take on single values in the context of each measurement.

3. Choose the dimensions – cont.

- Examples of common dimensions include:
 - date,
 - product,
 - customer,
 - transaction type,
 - status.

4. Identify the facts

- Facts are determined by answering the question, “What are we measuring?”
- All candidate facts in a design must be true to the grain defined in step 2.
- Facts that clearly belong to a different grain must be in a separate fact table.
- Typical facts are numeric additive figures such as quantity ordered or dollar cost amount.

Retail Case Study

- Let us consider a large grocery chain with 100 grocery stores spread over a five-state area.
- Each store has a full complement of departments, including grocery, frozen foods, dairy, meat, produce, bakery, floral, and health/beauty aids.
- Each store has roughly 60,000 individual products on its shelves.

Retail Case Study

- The individual products are called stock keeping units (SKUs).
- About 55,000 of the SKUs come from outside manufacturers and have bar codes imprinted on the product package. These bar codes are called universal product codes (UPCs).
- UPCs are at the same grain as individual SKUs.
- Each different package variation of a product has a separate UPC and hence is a separate SKU.

Retail Case Study

- The remaining 5,000 SKUs come from departments such as meat, produce, bakery, or floral.
- While these products don't have nationally recognized UPCs, the grocery chain assigns SKU numbers to them.
- Since our grocery chain is highly automated, we stick scanner labels on many of the items in these other departments.

Retail Case Study

- Although the bar codes are not UPCs, they are certainly SKU numbers.
- Data is collected at several places in a grocery store. Some of the most useful data is collected at the cash registers as customers purchase products.
- The POS system is at the front door of the grocery store where consumer takeaway is measured.
- The back door, where vendors make deliveries, is another interesting data-collection point.

Retail Case Study

- Both store management and headquarters marketing spend a great deal of time tinkering with pricing and promotions.
- Promotions in a grocery store include temporary price reductions, ads in newspapers and newspaper inserts, displays in the grocery store (including end-aisle displays), and coupons.

Retail Case Study

- The most direct and effective way to create a surge in the volume of product sold is to lower the price dramatically.
 - A 50-cent reduction in the price of paper towels, especially when coupled with an ad and display, can cause the sale of the paper towels to jump by a factor of 10.
 - Unfortunately, such a big price reduction usually is not sustainable because the towels probably are being sold at a loss.
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- Based on this business case study, we'll begin to design the dimensional model.

Step 1. Select the Business Process

- The first step in the design is to decide what business process(es) to model by combining an understanding of the business requirements with an understanding of the available data.
- The first dimensional model built should be the one with the most impact—it should answer the most pressing business questions and be readily accessible for data extraction.

Step 1. Select the Business Process

- In retail case study, management wants to better understand customer purchases as captured by the POS system.
- Thus the business process we're going to model is POS retail sales.
- This data will allow us to analyze what products are selling in which stores on what days under what promotional conditions.

Step 2. Declare the Grain

- Preferably you should develop dimensional models for the most **atomic** information captured by a business process.
- Atomic data is the most detailed information collected; **such data cannot be subdivided further.**
- Atomic data is highly dimensional – so is a perfect match for the dimensional approach.
- The less granular model is vulnerable to unexpected user requests to drill down into the details.

Step 2. Declare the Grain

- Aggregated summary data plays an important role as a performance-tuning tool, but it is not a substitute for giving users access to the lowest-level details.
- A data warehouse almost always demands data expressed at the **lowest possible grain** of each dimension not because queries want to see individual low-level rows, but because queries need to cut through the details in very precise ways.
- In our case study, the most granular data is an individual line item on a POS transaction.

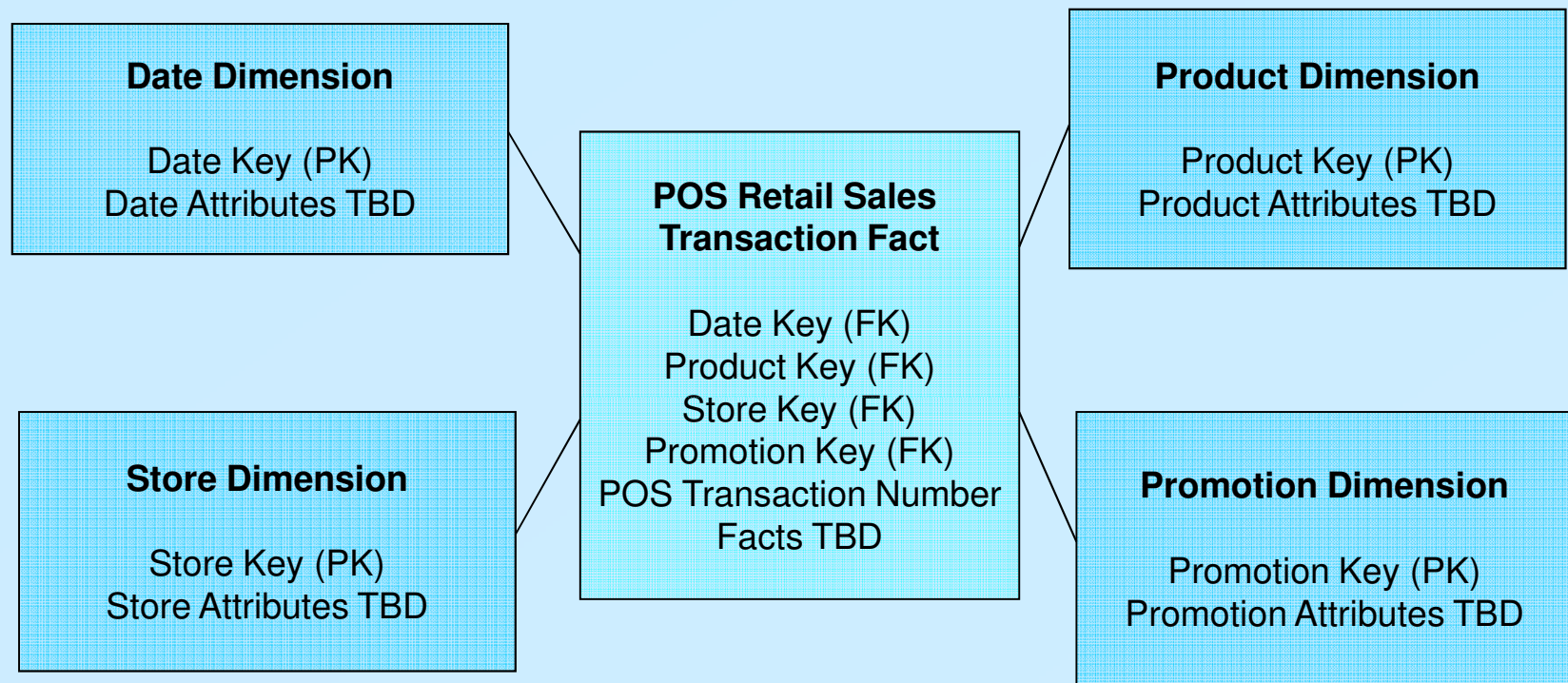
Step 3. Choose the Dimensions

- A careful grain statement determines the primary dimensionality of the fact table.
- It is then often possible to add more dimensions to the basic grain of the fact table, where these additional dimensions naturally take on only one value under each combination of the primary dimensions.
- If the additional dimension violates the grain by causing additional fact rows to be generated, then the grain statement must be revised to accommodate this dimension.

Step 3. Choose the Dimensions

- In our case study we identify 3 primary dimensions: the date, product, and store dimensions.
- Additionally we can add new dimensions, as the promotion under which the product is sold.
- In our case study we decide on the following descriptive dimensions: date, product, store, and promotion.
- In addition, we'll include the POS transaction ticket number as a special dimension (described later)

Step 3. Choose the Dimensions

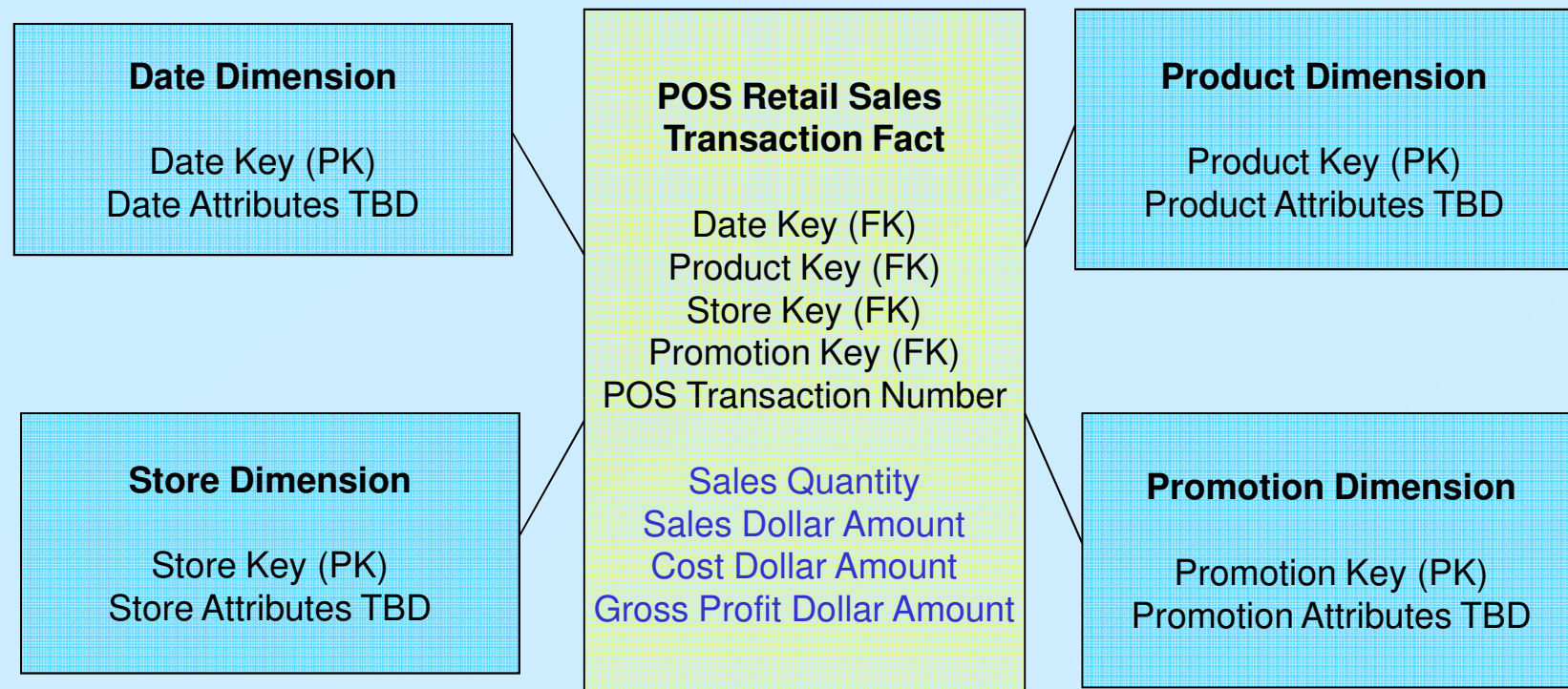


- TBD – To be determined

Step 4. Identify the Facts

- The facts must be true to the grain (in this case the individual line item on the POS transaction).
- The facts collected by the POS system include the sales quantity, per unit sales price and the sales dollar amount (=sales quantity * unit price).
- Some POS systems provide a standard dollar cost for the product as delivered to the store by the vendor.

Step 4. Identify the Facts



- TBD – To be determined

Step 4. Identify the Facts

- The **gross profit** is obtained by subtracting the cost dollar amount from the sales dollar amount.
- All these facts (sales quantity, sales dollar amount, cost dollar amount and gross profit dollar amount) are additive across all the dimensions.
- Percentages and ratios, such as gross margin, are nonadditive (gross margin = gross profit / dollar revenue)
- The numerator and denominator should be stored in the fact table.
- The ratio can be calculated in a data access tool for any slice of the fact table by remembering to calculate the ratio of the sums, not the sum of the ratios.
- Unit price is also a nonadditive fact (is a ratio)

Dimension Table Attributes

- The **date** dimension is the one dimension nearly guaranteed to be in every data mart because virtually every data mart is a time series.
- The time-of-day dimension is different from the date dimension.

Date dimension: Partial list of columns

Date Key (PK)	Calendar Quarter
Date	Calendar Year-Quarter
Full Date Description	Calendar Half Year
Day of Week	Calendar Year
Day Number in Epoch	Fiscal Week
Week Number in Epoch	Fiscal Week Number in Year
Month Number in Epoch	Fiscal Month
Day Number in Calendar Month	Fiscal Month Number in Year
Day Number in Calendar Year	Fiscal Year-Month
Day Number in Fiscal Month	Fiscal Quarter
Day Number in Fiscal Year	Fiscal Year-Quarter
Last Day in Week Indicator	Fiscal Half Year
Last Day in Month Indicator	Fiscal Year
Calendar Week Ending Date	Holiday Indicator
Calendar Week Number in Year	Weekday Indicator
Calendar Month Name	Selling Season
Calendar Month Number in Year	Major Event
Calendar Year-Month (YYYY-MM)	SQL Date Stamp

Date dimension – cont.

- The holiday indicator takes on the values of Holiday or Non-holiday.
- Weekday indicator have a value of Weekday or Weekend.
- The selling season column is set to the name of the retailing season, if any. Examples in the United States could include Christmas, Thanksgiving, Easter, Valentine's Day, Fourth of July, or None.
- The major event column is similar to the season column and can be used to mark special outside events such as Super Bowl Sunday or Labor Strike.

Date dimension – cont.

- Data warehouses always need an explicit date dimension table.
- There are many date attributes not supported by the SQL date function, including fiscal periods, seasons, holidays, and weekends.
- Rather than attempting to determine these nonstandard calendar calculations in a query, we should look them up in a date dimension table.
- Date and time are almost completely independent.
- If we combined the two dimensions, the date dimension would grow significantly.

Date dimension - Example

KeyDate	Date	Full Date Description	Day of Week	Calendar Month	Calendar Year	Fiscal Year-Month	Holiday Indicator	Weekday Indicator
1	01.01.2002	January 1, 2002	Tuesday	January	2002	F2002-01	Holiday	Weekday
2	01.02.2002	January 2, 2002	Wednesday	January	2002	F2002-01	Non-Holiday	Weekday
3	01.03.2002	January 3, 2002	Thursday	January	2002	F2002-01	Non-Holiday	Weekday
4	01.04.2002	January 4, 2002	Friday	January	2002	F2002-01	Non-Holiday	Weekday
5	01.05.2002	January 5, 2002	Saturday	January	2002	F2002-01	Non-Holiday	Weekend
6	01.06.2002	January 6, 2002	Sunday	January	2002	F2002-01	Non-Holiday	Weekend
7	01.07.2002	January 7, 2002	Monday	January	2002	F2002-01	Non-Holiday	Weekday
8	01.08.2002	January 8, 2002	Tuesday	January	2002	F2002-01	Non-Holiday	Weekday

Product Dimension

- The product dimension describes every SKU in the grocery store.
- A typical store in the chain may stock 60,000 SKUs, but when we account for different merchandising schemes across the chain and historical products that are no longer available, the product dimension would have at least 150,000 rows and perhaps as many as a million rows.
- The product dimension is almost always sourced from the operational product master file.

Product dimension: Partial list of columns

- Product Key (PK)
- Product Description
- SKU Number (Natural Key)
- Brand Description
- Category Description
- Department Description
- Package Type Description
- Package Size
- Fat Content
- Diet Type
- Weight
- Weight Units of Measure
- Storage Type
- Shelf Life Type
- Shelf Width
- Shelf Height
- Shelf Depth

Product dimension: Example

Product Key	Product Description	Brand Description	Category Description	Department Description	Fat Content
1	Baked Well Light Sourdough Fresh Bread	Baked Well	Bread	Bakery	Reduced Fat
2	Fluffy Sliced Whole Wheat	Fluffy	Bread	Bakery	Regular Fat
3	Fluffy Light Sliced Whole Wheat	Fluffy	Bread	Bakery	Reduced Fat
4	Fat Free Mini Cinnamon Rolls	Light	Sweeten Bread	Bakery	Non-Fat
5	Diet Lovers Vanilla 2 Gallon	Coldpack	Frozen Dessert	Frozen Foods	Non-Fat
6	Light and Creamy Butter Pecan 1 Pint	Freshlike	Frozen Dessert	Frozen Foods	Reduced Fat
7	Chocolate Lovers 1/2 Gallon	Frigid	Frozen Dessert	Frozen Foods	Regular Fat
8	Strawberry Ice Creamy 1 Pint	Icy	Frozen Dessert	Frozen Foods	Regular Fat
9	Icy Ice Cream Sandwiches	Icy	Frozen Dessert	Frozen Foods	Regular Fat

Product dimension – cont.

- A reasonable product dimension table would have 50 or more descriptive attributes.
- Each attribute is a rich source for constraining and constructing row headers.
- Viewed in this manner, we see that drilling down is nothing more than asking for a row header that provides more information.
- Drilling down in a data mart is nothing more than adding row headers from the dimension tables.
- Drilling up is removing row headers.
- We can drill down or up on attributes from more than one explicit hierarchy and with attributes that are part of no hierarchy.

Store Dimension

- The store dimension describes every store in our grocery chain.
- The store dimension is the primary geographic dimension in our case study.
- Each store can be thought of as a location. Because of this, we can roll stores up to any geographic attribute, such as ZIP code, county, and state in the United States.
- Stores usually also roll up to store districts and regions.
- These two different hierarchies are both easily represented in the store dimension because both the geographic and store regional hierarchies are well defined for a single store row.
- It is not uncommon to represent multiple hierarchies in a dimension table. Ideally, the attribute names and values should be unique across the multiple hierarchies.

Store Dimension: Partial list of columns

- Store Name
- Store Number (Natural Key)
- Store Street Address
- Store City
- Store County
- Store State
- Store Zip Code
- Store Manager
- Store District
- Store Region
- Floor Plan Type
- Photo Processing Type
- Financial Service Type
- Selling Square Footage
- Total Square Footage
- First Open Date
- Last Remodel Date

Promotion Dimension

- The promotion dimension describes the promotion conditions under which a product was sold.
- Promotion conditions include temporary price reductions, end-aisle displays, newspaper ads, and coupons.
- This dimension is often called a causal dimension (as opposed to a casual dimension) because it describes factors thought to cause a change in product sales.

Promotion Dimension - columns

- Promotion Key (PK)
- Promotion Name
- Price Reduction Type
- Promotion Media Type
- Ad Type
- Display Type
- Coupon Type
- Ad Media Name
- Display Provider
- Promotion Cost
- Promotion Begin Date
- Promotion End Date

Transaction Number: Degenerate Dimension

- Operational control numbers such as order numbers, invoice numbers, and bill-oflading numbers usually give rise to empty dimensions and are represented as degenerate dimensions (that is, dimension keys without corresponding dimension tables) in fact tables where the grain of the table is the document itself or a line item in the document.

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